PROJECT REVIEW 3

Final Demonstration, Evaluation and Project Completion Review

Maximum Marks: 10 | Team Size: 4 Students

Course Code / Title	
Team Number	
Project ID and Title	
Review Date	
Faculty / Evaluator	

Purpose: To evaluate the completed project through a reproducible live demonstration, verify correctness and technical depth, assess results and limitations, examine final documentation and repository quality, and award individual marks based on verified contribution and viva performance.

1. Purpose and Expected Outcome of Review 3

Review 3 is the final project evaluation. The team must present a complete and integrated system that satisfies the approved objectives or clearly justify any approved deviation. The evaluator should be able to understand the problem, execute the project, observe representative outputs, verify test evidence, and identify the contribution made by every student.

- Closure of all mandatory Review 2 corrections and completion of the agreed project scope.
- A stable, integrated and reproducible implementation with clear build and execution instructions.
- Live demonstration using representative valid, invalid, boundary and failure inputs.
- Evidence of correctness, comparison, benchmarking, performance analysis or other project-relevant evaluation.
- Clear explanation of algorithms, design decisions, compiler concepts, limitations and trade-offs.
- Complete source-code repository, README, test suite, final report and presentation.
- Verifiable individual contribution and the ability of each student to answer technical viva questions.
- Discussion of limitations, lessons learned, extension possibilities and publication or further-development potential.

2. Review 3 Evaluation Components – 10 Marks

Important: Final marks need not be identical for all team members. Individual marks will reflect demonstrated technical contribution, ownership of assigned modules, understanding of the complete system, ability to execute and explain the work, and performance in the viva.

Component	Evidence to Be Presented	Marks
1. Final System Completeness and Correctness	Completion of approved objectives; integrated operation; correct handling of required features; stability; meaningful error handling; reproducible execution.	2.5
2. Live Demonstration and Use-Case Coverage	Structured live demo; representative normal, invalid, boundary and failure inputs; clear explanation of input-output behaviour and module interaction.	1.5
3. Testing, Results and Evaluation	Test-suite summary; expected versus actual output; correctness evidence; comparison, benchmarking or performance metrics; interpretation of results.	1.5
4. Technical Depth, Design Decisions and Innovation	Appropriate compiler concepts and algorithms; complexity or trade-off discussion; justification of design; extensions, innovation or effort beyond the minimum scope.	1.0

Component	Evidence to Be Presented	Marks
5. Final Report, Repository and Reproducibility	Complete report; organised repository; README/build instructions; dependency information; meaningful commit history; documented tests, limitations and references.	1.0
6. Verified Individual Contribution	Member-specific technical ownership; commits/files/test artefacts; ability to demonstrate the assigned work; contribution log and integration responsibility.	1.5
7. Presentation and Viva	Clarity, time management, professional presentation, understanding of the complete system, response to questions, ethical acknowledgement of reused resources.	1.0
Total		**10.0**

3. Member-Wise Presentation Breakdown for a Team of Four

Recommended total presentation and demonstration duration: 18–22 minutes, followed by 8–12 minutes of viva. The live system demonstration should occupy at least 8 minutes.

Presentation rule: Every member must demonstrate a technical part of the completed project and explain the corresponding source code, tests and results. Merely presenting the introduction, conclusion, slides or report does not establish equal technical contribution.

Presenter	Recommended Time	Content to Be Presented	Mandatory Personal Evidence
Member 1 – Project Lead / System Integrator	4 minutes	Introduce the completed problem and objectives; explain final architecture, changes from the approved plan, overall completion status, integrated workflow and major design decisions.	Final objective-compliance matrix, integrated build, architecture, integration commits and closure of review action items.
Member 2 – Front-End / Analysis Module Owner	4–5 minutes	Explain and demonstrate the assigned front-end, parsing, semantic, data acquisition, analysis, type-checking or equivalent module, including important algorithms and error cases.	Source files, commit history, module tests, execution trace, sample valid/invalid inputs and fixes completed.
Member 3 – Core Processing / Back-End Owner	4–5 minutes	Explain and demonstrate the assigned optimization, IR, code-generation, VM/runtime, ML, security, visualisation or other core processing module and its interaction with the system.	Source files, commits, transformation/output examples, performance or correctness results and module limitations.
Member 4 – Testing / Evaluation / Release Owner	4–5 minutes	Demonstrate the final test suite, results, comparisons or benchmarks; explain error handling, packaging, README, reproducibility, limitations, documentation and future extension.	Automated/manual test evidence, result tables, charts/logs, issue closure, final repository and report contributions.
All Members – Integrated Demo and Viva	8–12 minutes	Execute the complete project, explain module transitions, handle evaluator-selected inputs, identify source-code locations, justify design choices and discuss limitations and extension potential.	Individual ability to operate the system, explain own and connected modules, interpret output and respond independently.

4. Recommended Review 3 Presentation Structure

Recommended slide limit: 14–18 content slides. The presentation must support the live demonstration, not replace it. Keep source-code excerpts, additional test evidence, algorithm traces, comparison tables and references as backup slides for viva.

Slide	Required Content	Primary Presenter
1	Project title, project ID, team number and member details	Member 1
2	Problem, motivation and final measurable objectives	Member 1
3	Review 2 feedback closure and final scope achieved	Member 1
4	Final architecture, modules, interfaces and execution workflow	Member 1
5	Key design decisions, algorithms and compiler concepts used	Members 1–3
6	Module 1 design, implementation and important source-code logic	Member 2

Slide	Required Content	Primary Presenter
7	Module 1 demonstration with valid and error inputs	Member 2
8	Module 2 design, implementation and important source-code logic	Member 3
9	Module 2 demonstration and integration with the complete system	Member 3
10	Complete end-to-end live demonstration using representative use cases	All Members
11	Test strategy and test-suite coverage	Member 4
12	Results, correctness evidence, comparison or benchmarking	Member 4
13	Observed limitations, failure cases and technical trade-offs	Members 3 & 4
14	Repository, build/run instructions and reproducibility	Member 4
15	Member-wise technical contribution and integration responsibility	All Members
16	Innovation, extensions and publication/future-development potential	Member 1
17	Conclusion: objectives achieved and key learning	All Members
18	References and acknowledgement of tools/libraries/resources	Member 4

5. Final Project Document and Submission Package

Each team should submit a consolidated final report and complete project package. Recommended final-report length: 15–25 pages excluding appendices, source-code listings and generated test logs. The report should reflect the actual implementation and results demonstrated during Review 3.

Cover Page and Declaration: Course, project ID/title, team number, members/register numbers, review date, faculty and originality declaration.

1. Abstract: Problem, approach, implementation and principal results (200–300 words).
2. Introduction and Motivation: Context, need, beneficiaries and relevance to compiler design.
3. Problem Statement, Objectives, Scope and Limitations: Final approved scope and measurable objective-compliance table.
4. Background / Related Work: Concepts, prior tools or research, comparison and identified gap.
5. Requirements and Technology Stack: Software, hardware, libraries, datasets/grammars/test programs and environment.
6. System Design: Final architecture, data flow, modules, interfaces, data structures and design decisions.
7. Algorithms and Methodology: Parsing/IR/optimization/analysis/code-generation/runtime or project-specific methods with complexity and trade-offs.
8. Implementation Details: Module-wise explanation, important code organisation, build process and integration.
9. Testing Strategy and Test Cases: Positive, negative, boundary, stress and regression cases with expected and actual results.
10. Results and Evaluation: Correctness, comparison, performance, visualisations, benchmark setup and interpretation.
11. Error Handling, Limitations and Known Issues: Conditions not supported, technical constraints and impact.
12. Member-Wise Contribution: Technical ownership, commits/evidence, integration and testing contribution.
13. Conclusion and Future Work: Objectives achieved, learning, extension and publication/research potential.

14. References: Books, papers, official documentation, datasets, libraries and reused open-source components.

Appendices: User manual, installation guide, sample runs, complete test table, API/grammar specification, contribution log and additional diagrams.

6. Mandatory Final Contribution Matrix

Member	Primary Technical Ownership	Final Deliverable Produced	Evidence / Commit / Test	Integration / Support Contribution	Viva Module
Member 1					
Member 2					
Member 3					
Member 4					

Guideline: Each member must own a measurable technical output. The evaluator may inspect repository history, source files, test artefacts, diagrams, issue records or execution evidence to confirm the contribution claimed in this matrix.

7. Mandatory Final Submission and Demonstration Checklist

☐ Final presentation following the prescribed structure.

☐ Complete final report in the required format.

☐ Stable integrated source code and executable/buildable project package.

☐ Repository link with complete history, README, licence/acknowledgements and dependency details.

☐ Installation, build and run instructions verified on the demonstration system.

☐ Complete test suite with valid, invalid, boundary, failure and regression cases.

☐ Result tables, logs, charts, comparison or benchmarking evidence applicable to the project.

☐ Sample input files, grammars, datasets, programs or configuration needed for reproduction.

☐ Final architecture, module, interface and data-flow diagrams matching the implementation.

☐ Member-wise contribution matrix and contribution log supported by technical evidence.

☐ List of known limitations, unsupported cases and future enhancements.

☐ Backup copy of code, report, presentation, sample data and demonstration recording where permitted.

☐ All external code, libraries, datasets, diagrams and ideas properly acknowledged.

☐ No confidential credentials, personal data or unapproved copyrighted material included in the submission.

8. Suggested Final Demonstration Sequence

Stage	Demonstration Activity	Expected Evidence
1. Environment	Show repository/version, dependencies, build command and execution command.	Reproducible setup and successful build.
2. Normal Use Case	Run a representative valid input through the complete project.	Correct output and explanation of module flow.
3. Alternative / Boundary Case	Run a boundary or less-common valid case.	Robustness and correct handling of edge conditions.
4. Invalid / Failure Case	Provide invalid input or trigger an expected failure condition.	Meaningful detection, diagnostic or safe failure behaviour.

Stage	Demonstration Activity	Expected Evidence
5. Internal Evidence	Show selected source code, IR/AST/CFG/log/trace or intermediate representation.	Technical understanding of internal operation.
6. Evaluation Result	Show test summary, comparison, metric, benchmark or correctness result.	Evidence supporting project claims.
7. Evaluator-Selected Input	Execute an input or scenario selected during viva.	Authenticity, readiness and understanding.

8.1 Evaluator-Selected Test Record

During the final viva, the evaluator may provide unseen or modified inputs. Record the expected behaviour, observed output and result below. This table may also be used to document reproducibility on the evaluation system.

Test / Scenario	Expected Behaviour	Observed Output / Evidence	Result / Evaluator Remarks
1. Evaluator-selected input			
2. Boundary or failure condition			
3. Rebuild / reproducibility check			

9. Review 3 Evaluator Record

Criterion	Maximum	Marks Awarded	Evaluator Remarks
Final system completeness and correctness	2.5		
Live demonstration and use-case coverage	1.5		
Testing, results and evaluation	1.5		
Technical depth, design decisions and innovation	1.0		
Final report, repository and reproducibility	1.0		
Verified individual contribution	1.5		
Presentation and viva	1.0		
Total	**10.0**		

Overall Final Decision: ☐ Completed satisfactorily ☐ Completed with minor corrections ☐ Incomplete / major correction required

Final Remarks / Corrections / Distinctive Achievement:

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Team Leader Signature	Faculty / Evaluator Signature	Date